CHALMERS FORMULA STUDENT

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Autonomous Hardware and Telemetry
Final Report– CFS24

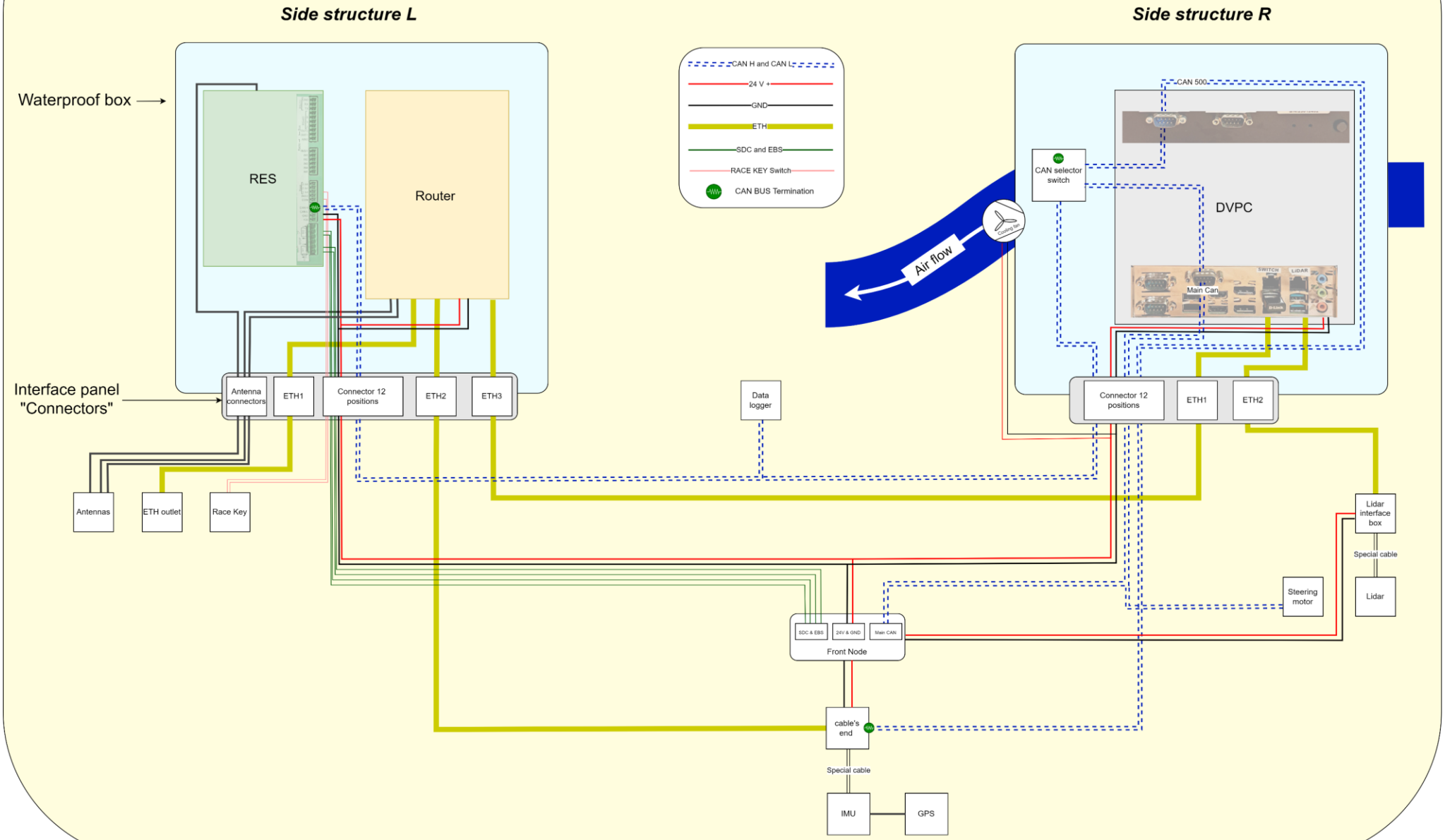
Background

The vehicle needs to perform calculations based on many inputs in both manual and autonomous modes. To achieve that, the vehicle will be equipped with devices that sense the environment (sensores), followed by implementing some algorithms, and lastly requesting the decided actions from the other subsystems.

Most of the communication between the subsystems in the car uses CAN buses. That is due to its properties in discarding the noises. A few subsystems where the data transfer rate is high the CAN bus will be replaced by ethernet.

System Overview

Two sidepods solution



System overview

The system overview shows the wiring diagram of all the components I am working on. The two blue boxes are waterproof. They will be placed inside both side structures. One box each.

The boxes will include connectors mounted to the box side. The devices inside the box will be connected to these connectors. The other side of the connectors will be connected to the other subsystems in the vehicle.

This solution will provide modularity. The devices still be detachable and even the whole box could be built in our workshop, SR, and when it is done can be mounted to the car.

All connectors will be waterproof. The types will be ethernet cable carriers which will allow us to keep the standards. A Superseal or a similar connector will be needed for the powering and other wires. RP-SMA antenna connectors and SMA antenna connectors.

More in depth RES

According to the rules, a formula student car must be equipped with an RES if it participating in driverless events. The RES can not be a part of our telemetry. The rules specify which RES devices should we have and how it needs to be connected and configured.

The RES comes in two parts. A vehicle module (receiver) and a transmitter module. The transmitter uses a Race E-key which specifies which frequencies the vehicle module will be operated at.

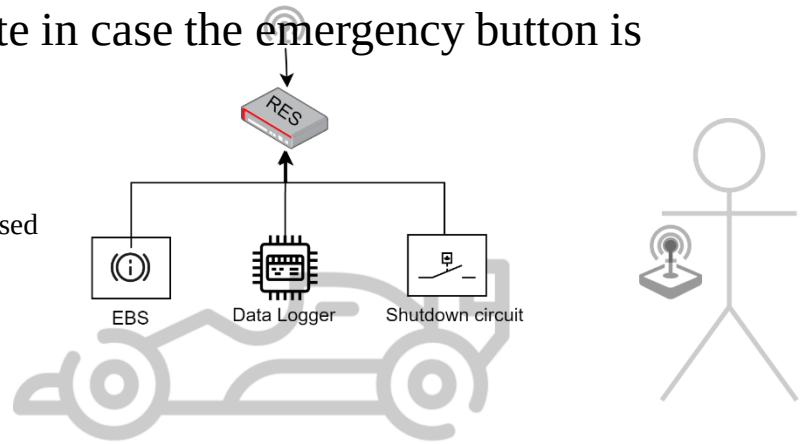
The RES needs to be connected to a CAN bus so the data logger can catch the RES messages.

The RES has a Race key switch, which lets the RES operate at the correct frequencies based on the Race E-key. According to FSG's new handbook, the Race key switch has to be located close the dash.

The RES must also be a part of the SDC.

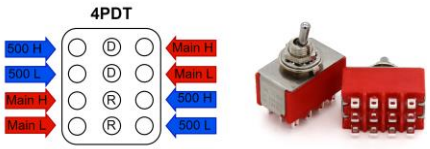
The RES must also be connected to the EBS to activate in case the emergency button is pressed on the transmitting module.

- When the Shut Down Circuit, SDC, triggers it disconnect the high voltage circuit.
- When the Emergency Breaking System, EBS, triggers it engage the breakings calipers used pressurized air.



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CAN Switch Selector



Labels
- MAIN CAN
- RES CAN
- IMU CAN
- PE CAN

Currently used

4PDT

TSAC

AMS

PE

RN

RES

Left Side structure

HVD

CDL

FSEast Transponder

CAM

RES CAN

IMU

IMU CAN

4PDT

PC

Right Side structure

Main CAN

FN

DASH

STA

Kvaser

Main CAN

More in depth CAN bus

This year we introduced the CAN switch selector which is a 4PDT toggle switch. The competition datalogger (CDL) and the RES need to be on the same CAN bus and the dataloggers are different for different competitions therefore their CAN bus compatibility is different. As a result, we equipped our car with two CAN buses that operate at different data rates to meet the two versions of the competition's data loggers.

Since switching the RES and CDL between the CAN was a bit sketchy in CFS23 in terms of CAN termination and wiring, we introduced the CAN switch selector.

We designed our car based on the 2023 formula student competition's handbooks. After implementing the CAN switch selector the FSE realsed a new version of their data logger that operate at the same rate as the FSG therefore the switch is not needed any more. We will still have the switch in the car because our car is still compatible the FS24 competitions that we will participate, and our car got the bonus to be able to operate the CDL at different data-rate if needed for the future.

https://fseast.eu/wp-content/uploads/2023/05/FS_EAST_2023_05_Data_logger_A4_v1.pdf

https://fseast.eu/wp-content/uploads/2024/04/FS_EAST_2024_04_Data_logger_A4_v1.pdf

More in depth IMU

- **IMU**: Inertia measurements unit is the device responsible for sensing the acceleration in all directions. It has an integrated GPS unit. But needs to be connected to a GPS antenna to get a higher GPS resolution. The IMU needs to be placed close to the vehicle's center of mass to return reliable measurements. The IMU we are using is very expensive. It is able to process and filter the raw data and send the orientation and acceleration estimations to the DVPC.

The IMU communicates with the DVPC through a CAN bus. The main CAN bus we have in the car is running at a speed of 1000kbit/s while the IMU is capable of being connected to a CAN bus with a speed of 500kbit/s or 1000kbit/s.

We choose to run it on a seperate @ 500kbit/s CAN bus due to:

- Not overcrowd the main CAN.
- Since the IMU data is only used in the DVPC.
- The old FSEast 2023 require an RES CAN bus at 500Kbit/s.

Since the CAN bus starts at the IMU, it needs to be terminated there. The problem is that the IMU needs a special cable and therefore the termination resistance will be placed at the end of that cable. The cable needed is around 20cm long and we will terminate the CAN bus at the end of this cable. (Result: The termination at the end of the cable after 20 cm worked fine.)

The IMU can be configured via an ethernet connection which is why it also needs to be accessible. The IMU is connect to the ethernet to the in-car router that acts as a network switch and therefore is accessible through the Ethernet outlet under the dashboard and even wireless through the telemetry.



More in depth Steering motor

The steering actuator is a motor connected to the steering shaft that will steer the vehicle in autonomous mode. It communicates with the DV PC via a CAN bus.

In CFS23 the motor responds to the CAN messages with a CAN ID = 0 which is also used in the RES. That is why the team run a separate CAN bus between the DVPC and the steering motor.

CFS24 started working on building a new steering motor from scratch but it is not expected to be done under CFS24. The team bought a new steering motor to have it in this year's car. The new motor has already been tested and therefore it will be connected to the main CAN.

More in depth LIDAR

LIDAR is the device that creates a 3D model of the environment surrounded by the vehicle. It helps the vehicle navigate in autonomous mode. The lidar we have, needs an interface box that will split the output of the lidar into power and data cables.

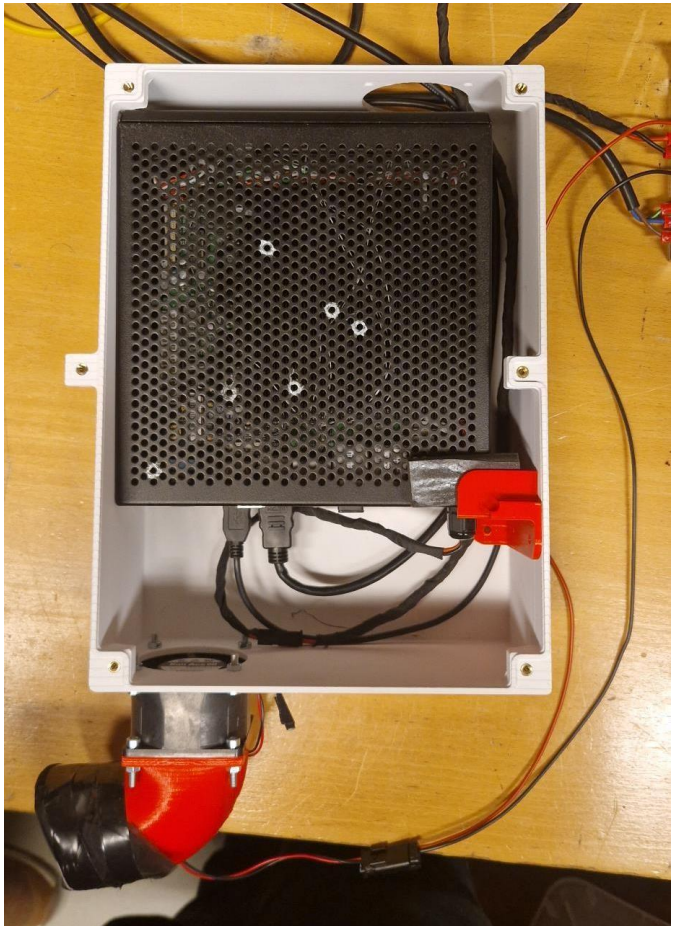
It communicates with the DVPC via an ethernet cable through the interface box. The ethernet cable runs between the side structure, where the DVPC is located, and the lidar mounted on top of the main hoop. This means this ethernet cable runs alongside the high-voltage cables for the FR motor, which is why it is important to use a shielded Ethernet cable for this task.

The Autonomous sub-group wants to have it in the nose of the monocoque as in CFS23. The new competition's rules say that the team needs to perform a crash test if the lidar will be placed in the front. The crash test is too expensive and needs a lot of preparation. We tested having the lidar on the main hoop and the autonomous subgroup accepted the result. By placing the lidar on the top of the main hoop some parts of the car will appear in the lidar's output which means losing a small amount of the lidar's field of view.

More in depth DVPC 1/2

The driverless computer is the brain of the vehicle. It will perform all calculations needed to run the car in both autonomous and manual mode.

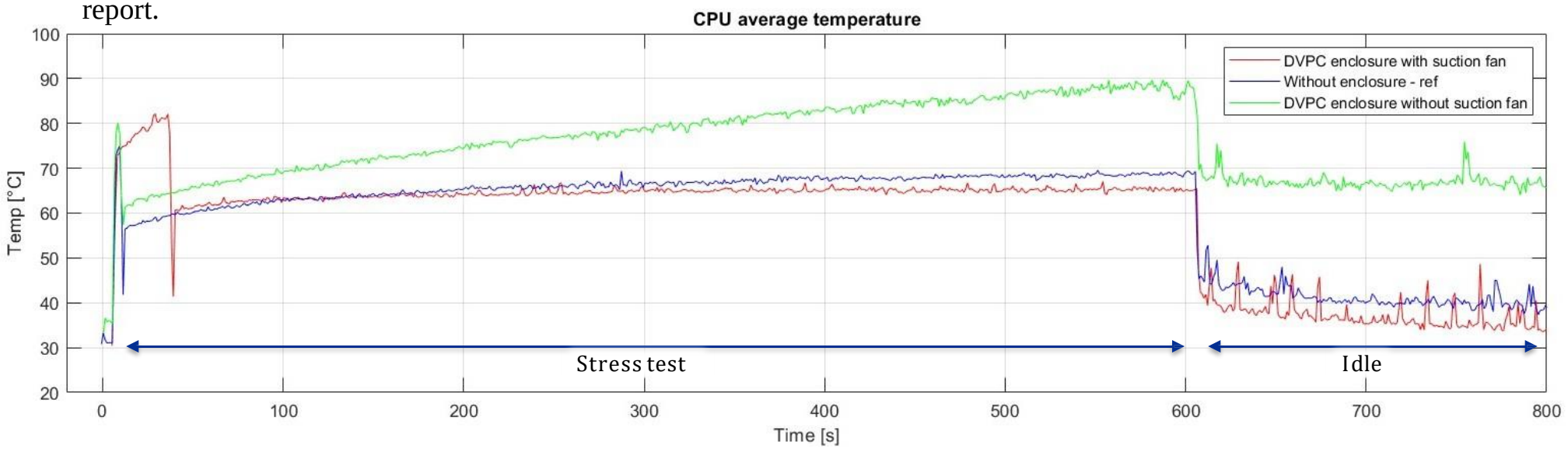
It is connected to the Lidar, CAN busses, and the in-car router. It consumes around 150 w and gets its power from the front node. It needs to be cooled since it is placed inside an enclosed space. The CPU heatsink has been upgraded from the default one that CFS23 had to get better heat dissipation. The waterproof enclosure is now equipped with a suction fan that will generate sufficient airflow around the CPU's heatsink. To validate the cooling design we conducted a stress test. The stress test pushes all the CPU's threads to their limit and the CPU's temperature and its clock-rate have been recorded. When stressing the CPU, the temperature increases immediately. After a couple of seconds the CPU throttles and lowers its clock rates to save the processor from getting damaged. The time needed for the CPU to throttle can not be used as a measure to validate the cooling performance. The throttling process is complicated and depends on many unknown parameters.



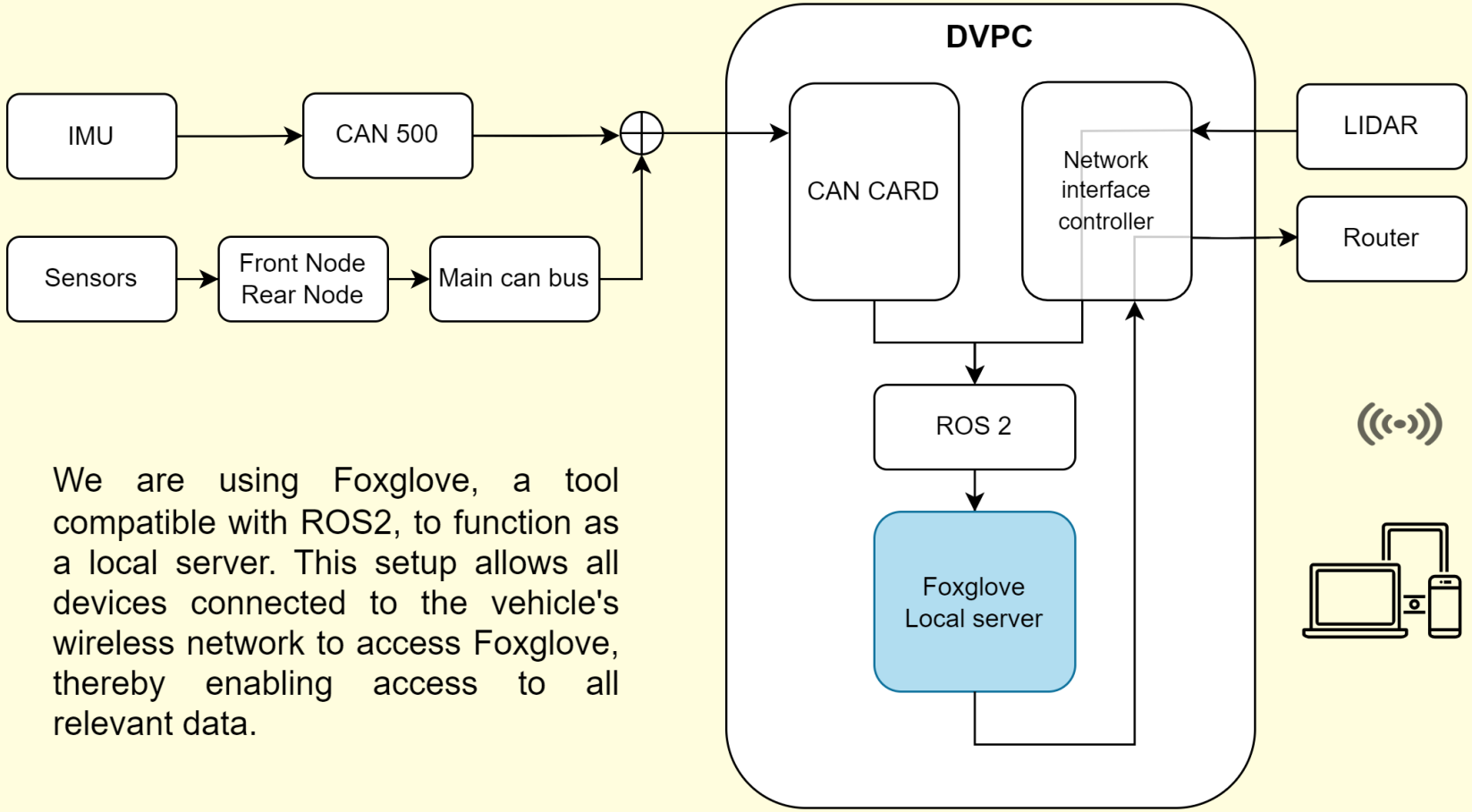
More in Depth DVPC 2/2

We did the same test three times.
- The PC was stressed outside the box.
- The PC was stressed inside the box when the suction fan was turned off and the air inlet was blocked.
- The PC was stressed inside the box and the suction fan was turned on while the inlet was covered with 4 layers of a hydrophobe mesh to simulate the airflow obstruction in the final setup.

The conclusion is the CPU is equally cold or even cooler when we have it inside the enclosure and the suction fan is on than running it outside the enclosure on the table in SR. More about DVPC cooling can be found in Adithya Sivakumar report.



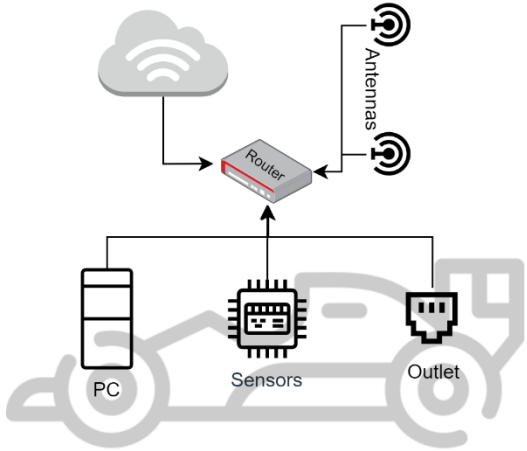
Telemetry Communication

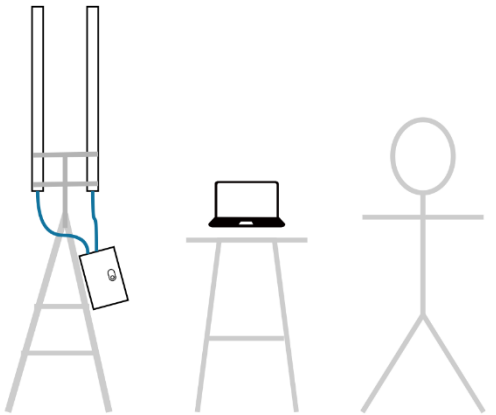


We are using Foxglove, a tool compatible with ROS2, to function as a local server. This setup allows all devices connected to the vehicle's wireless network to access Foxglove, thereby enabling access to all relevant data.

More in depth Telemetry 1/13

Our telemetry use case involves monitoring our racing car's status in real-time during competitions and testing. This can be accomplished by having a router in the car connected to the main PC and streaming a Wi-Fi signal that can be picked up by another router connected to a computer to visualize the data. We even added a 4G internet connection so in case we can not reach the car through the wifi we can still reach it via internet.





More in depth Telemetry 2/13

Telemetry parameters:

- *Real-time monitoring.*
- *Wifi 2.4 GHz.*
- *Access to the internet through 4G.*
- *Coverage radius: ≈ 400 m*
- *The racing/testing track is flat with minimal obstacles.*
- *Expected data transfer rate: ≈ 10Mbit/s*

More in depth Telemetry 3/13

Router selection:

- *The router needs to support 2.4 GHz.*
- *The router should be able to be powered by our LVS which is 24v.*
- *The router needs to be able to connect to external antennas.*
- *The router needs to include an internal network switch with 3 LAN ports.*
- *The router needs to be able to connect LTE/4G.*
- *The router needs to be Light weight.*
- *The router needs to be robust.*

We found that most of Teltonika's routers will fulfill our requirements and even they have developed a remote management system (RMS). More about RMS in the coming slides.

More in depth Telemetry 4/13

- Software and networking:

All control algorithms and autonomous systems are based on ROS2 which is already connected to the CAN buses and includes all the data that can be interested to be monitored through telemetry.

ROS2 works in node topology. The CFS23 team implemented a node in ROS2 that subscribes to all other nodes and publishes the values to a software called "FOXGLOVE" through a web socket. DVPC assigns a specific port to Foxglove. By knowing the local IP address of the DVPC and the foxglove's port, the data will be accessible from all other devices inside the local network.

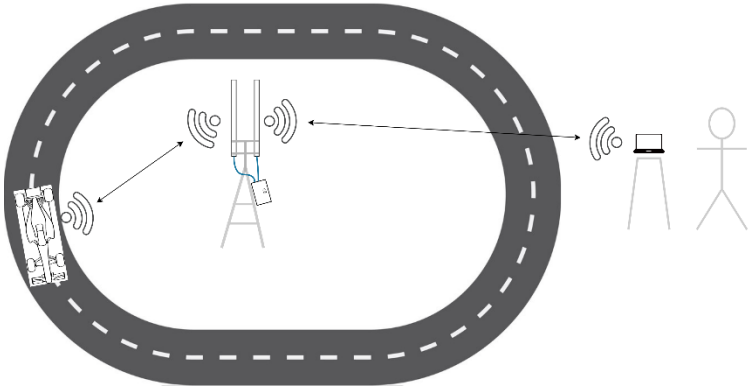
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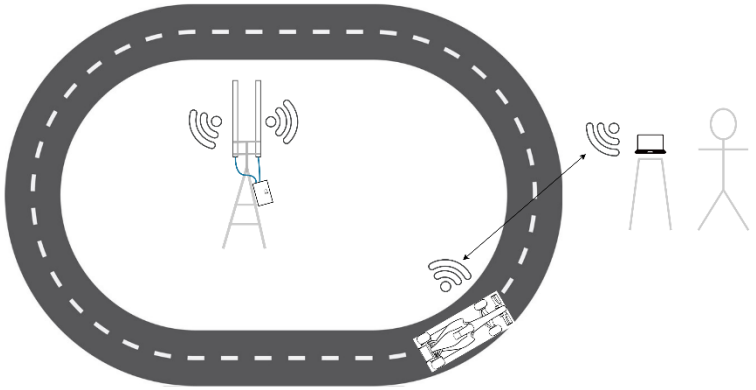
More in depth Telemetry 5/13

- Access the DVPC via WIFI

We need to connect the base station router to the in-car router.
One solution is to configure the base station router as a client to the on-car router. In this case, we know that the routers are communicating with each others via wifi 2.4 Ghz.
Another solution is that we can configure both routers as Mesh. This means that both routers are broadcasting the same WiFi network.
Mesh networking will allow us to place the mobile base station in the middle between the farthest place the car can reach from the team. this will ensure a that the recived signal strength is at least equal to the link budget between the base station and the team which is half of the original distance . And in best case scenario when the car get closer to the team than the base station the receiving device will seamlessly switch the connection the car directly to obtain a better link-budget.

More in depth Telemetry 6/13





- Worse case using mesh:

If the car is farther away than the base station, the team will reach the car through the base station

- Best case using mesh:

If the car get closer to the team, we can reach the car directly

Since we are using mesh mode, the transition between the worse case and the best case will be seamless.

More in depth Telemetry 7/13

- *Access the DVPC via 4G:*

To access the DVPC from the internet the DVPC needs to have an internet connection and needs to be assigned to a static IP address which is a bit complicated. Thanks to Teltonika RMS services we can access all the local IP:s in our local network from the internet using a virtual private network VPN.

By connecting the Teltonika router to the internet and registering the router's MAC-address to Teltonika RMS we will be able to download a custom-made VPN to our computers that are not connected to the DVPC's local network and still be able to access the DVPC through its local IP address through the internet.

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More in depth Telemetry 8/13

Usually, the wifi signal does not travel long due to multiple reasons. The most important one is the regulation in Europe that limits the wifi signal power to 20 dBm e.i.r.p at 2.4 GHz.

Teltonika's routers like all other license-free routers complies with the EU directive. To obtain a larger coverage area we will try to use our 20 dBm in the most efficient way.

The on-car router is placed inside the side structure, and we will run extension antenna cables between the router and the antennas. Introducing these extension cables will introduce signal losses. To compensate for the signal loss in the cables we will replace the included WiFi antennas with higher gain antennas and even boost the transmit power on the car side. The transmit power of the base station will be reduced to compensate for the high gain antennas. Higher gains antennas at the base station side are used to ensure a better receiving of weak signals which in our case extend the coverage area of our telemetry.

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More in depth Telemetry 9/13

Antennas characteristics:

- *Radiation pattern: Two omnidirectional antennas will be used to ensure good signal coverage around the car.*
- *Gain: This parameter shows how good is the antenna in terms of focusing the signal. Higher gain means a larger coverage area. We are using 2X 6dBi on the car and 2X 12 dBi on the base station.*
- *Type: Monopole/Dipole. Old Monopole antennas needed a ground plane for the antennas to work properly while dipole antennas do not require that. With the new antenna technology even monopole antennas that are printed on PCB and include waveguides do not need a ground plane anymore. We are using modern monopoles.*
- *Polarization: Vertical/horizontal polarization. The polarization of the transmitter (on car) and the receiver (base station) antennas must have the same polarization. We use vertical.*
- *Voltage standing wave ratio VSWR: This parameter shows the ratio between the transmitted/ received signal and the reflected signal inside the antenna. In other words, less VSWR means less return loss and a more efficient antenna.*

More in depth Telemetry 10/13

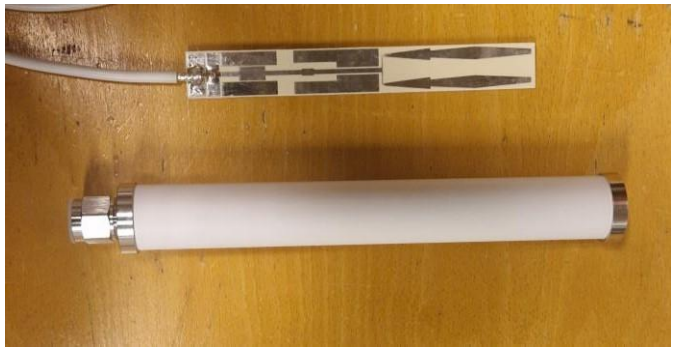
Antennas sponsorship from QuWireless:

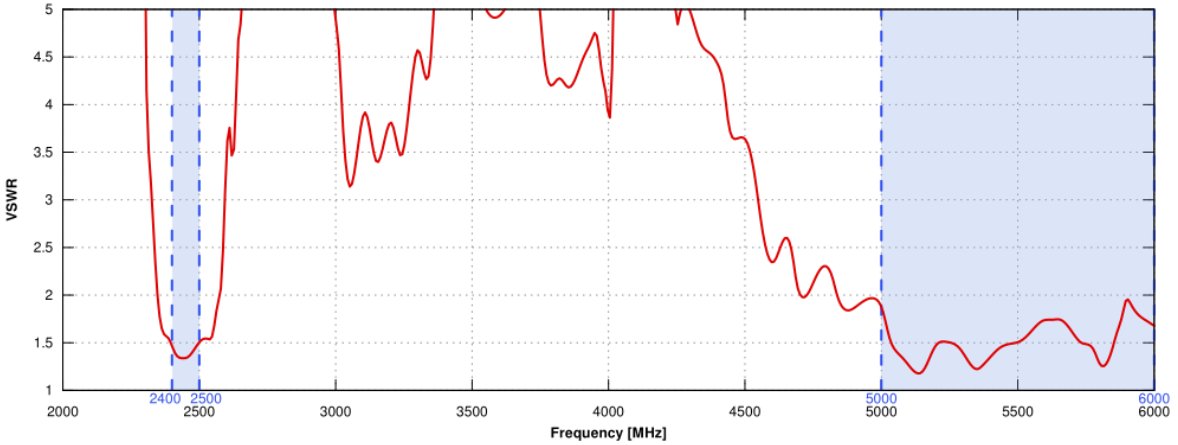
A company from Poland sponsored us with antennas and antenna cables.

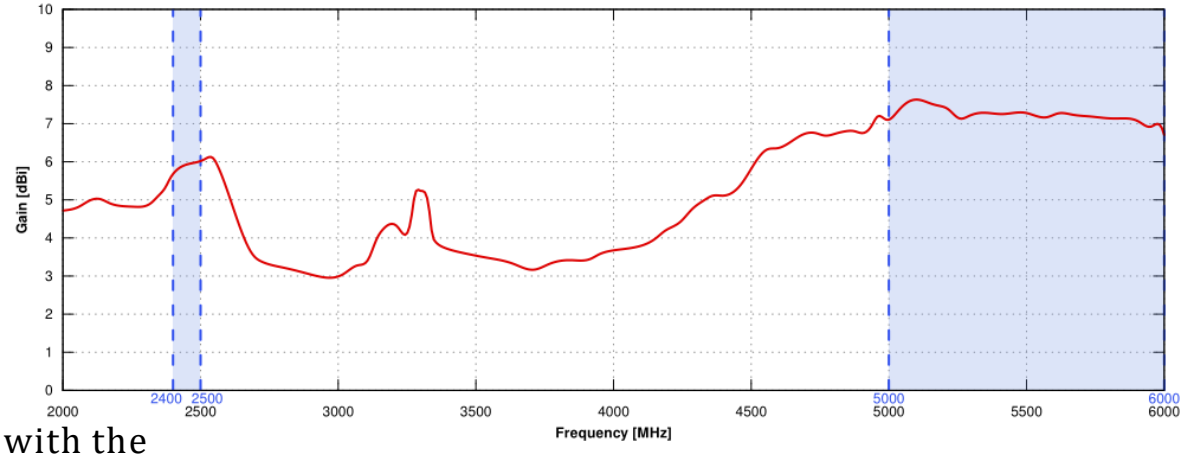
The company recommended using exposed antennas PCB on the car to save weight. And they provided us with 4 antennas so we can choose what we like.

- *2x 6 dBi Cylindrical omni antenna*
- *2x 6 dBi Omni antenna's PCB*

We choose to use the protected cylindrical antennas for better waterproofing and robustness.







6 dBi antennas. Cylindrical and exposed PCB with the same characteristics shown on the graphs to the right

More in depth Telemetry 11/13

- *Rules:*

EU: The wifi signal power e.i.r.p must be ≤:

20 dBm @ 2.4 GHz

23 dBm @ 5 Ghz

Competition: The whole antenna must be inside the surface envelope if it is longer than 10 cm.

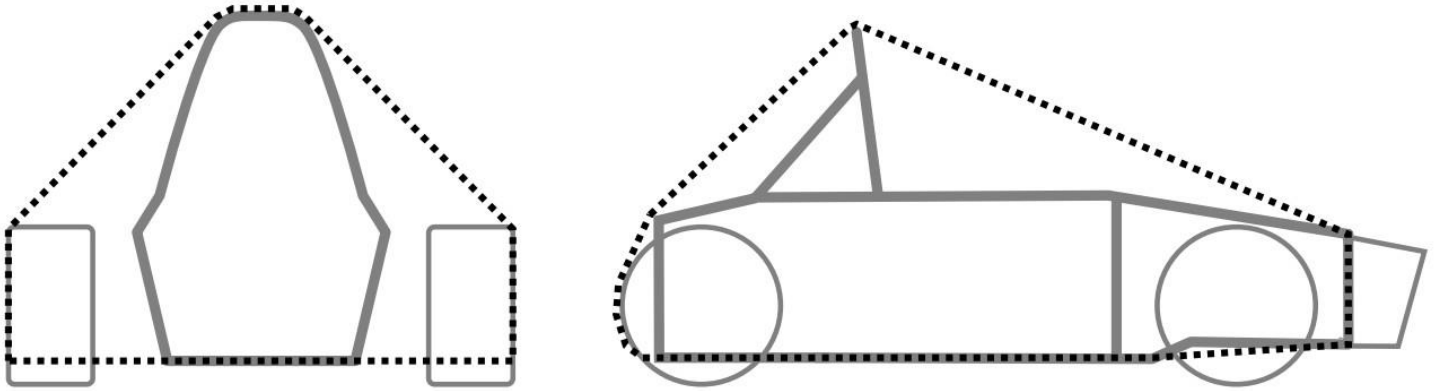
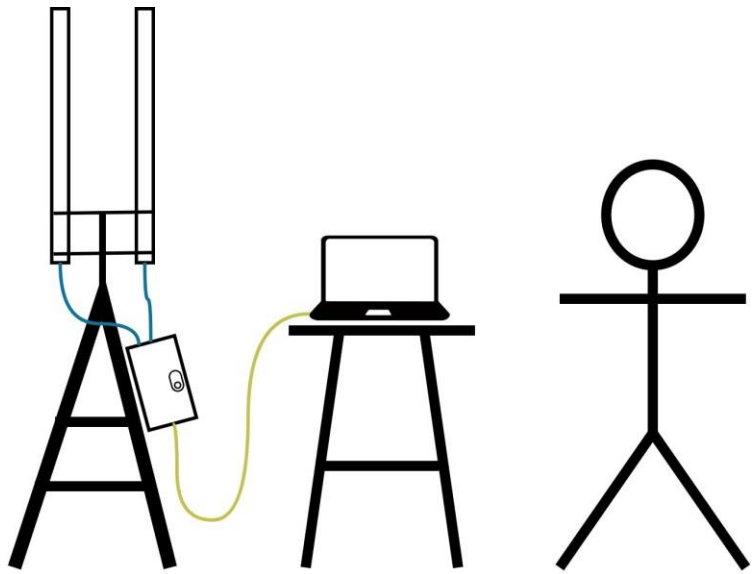


Figure: The surface envelope defined in the FSG rule book

More in depth Telemetry 12/13

Base station:

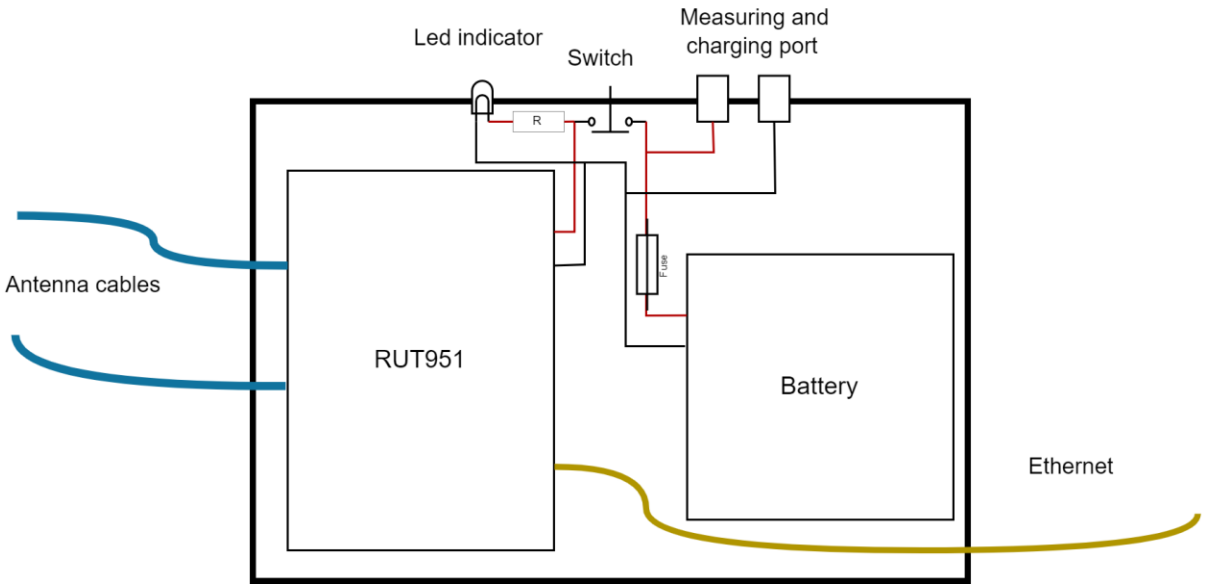
It will be equipped with RUT951 with two 12 dBi antennas *(sponsored by QuWirless)* *mounted on a tripod. It will be powered by battery to keep the system portable and due to competitions that do not allow bringing complex systems near the racing track.*



Rule D2.3.8: When driving in autonomous mode, an ASR has to be present at the race control with the RES. Additionally, one single monitoring device (laptop, tablet, …) may be brought (no complicated antenna construction or similar!)

More in depth Telemetry 13/13

Base station from the inside:



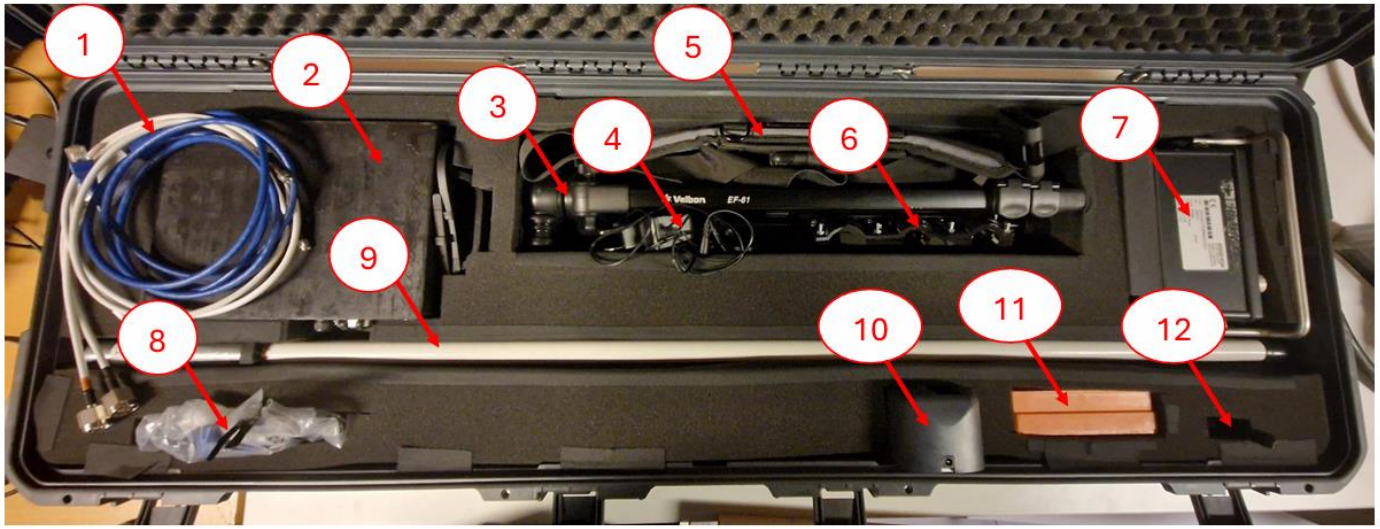
- *The base station can be charged by connecting the measuring points (banana sockets) to a power supply providing 14.6 V and a maximum current of 5 A. When the charging current drops below 70 mA, the power supply should be disconnected.*
- *I used a 10 A fuse to prevent damage to the base station in case of an incorrect charging process or short-circuiting of the measuring points.*
- *The LED indicator is quite bright because it needs to be visible during operation under direct sunlight at the competitions in summer.*
- *A fully charged battery can power the base station for 10 to 30 hours, depending on the number of connected devices and the network traffic.*

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Telemtry Hard Case

Packaging instructions and components' list

1- 2x **Antenna's cables**, **ethernet cable**, and ethernet to **usbc adapter**	7- **RES** transmitter module
2- Base station **electronics box**	8- **Tools' bag**: zip ties, Allen keys, cable cutter, electrical tape.
3- **Tripod**	9- 2x **Antennas**
4- RES charger **adapter**	10- RES **charger**
5- RES **belt**	11- 2x RES **batteries**
6- 3x Antenna **mounts**	12- Long **RES antenna** and a short RES antenna



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Telemetry: Test ideas

Telemetry coverage test

- *This will help us to know how good is our system and what should we expect during car testing and competitions.*

Using a VNA (Vector network analyzer)

- ***Generate VSWR graphs (2-6 GHz) for the antennas at different placement on the car. Using both the complete antennas and the PCB's antenna.***
  - *This will help us understand the impact of having different materials surrounding the antennas.*
  - *This will assist us to finalize the antenna's placement.*

Using a spectrum analyzer (requires also an antenna chamber)

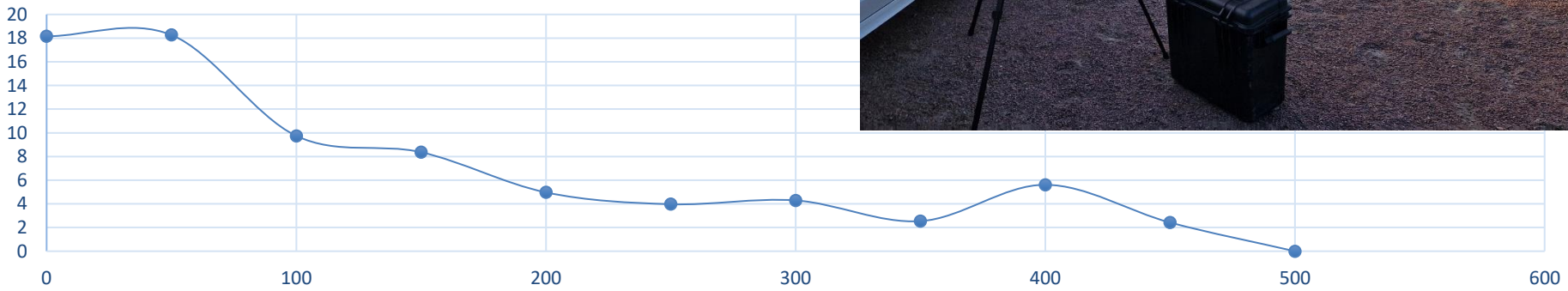
- ***Generate a gain graph (radiation pattern) for the antenna in different antenna placements. Both for 5 and 2.4 GHz.***
  - *This will help us to understand how efficient are the antennas in focusing the beam.*
- ***Measuring the Equivalent isotropic radiated power (E.I.R.P) for both car's antennas and the base station's antennas.***
  - *To stay within the legal limit.*
  - *To estimate the coverage radius using the FSPL equation.*

Telemetry: Test results

Coverage test results

A coverage test has been conducted on an open agricultural field. Where one of the car antennas was mounted close to the ground in front of a test-panel that could represent the effect of having the antennas close to the monocoque on the car.





Y axis shows the data rate in [Mbit/s] while the x-axis is the distance from the base-station in [m].

Telemetry: Test results

- *EMC Services AB:*

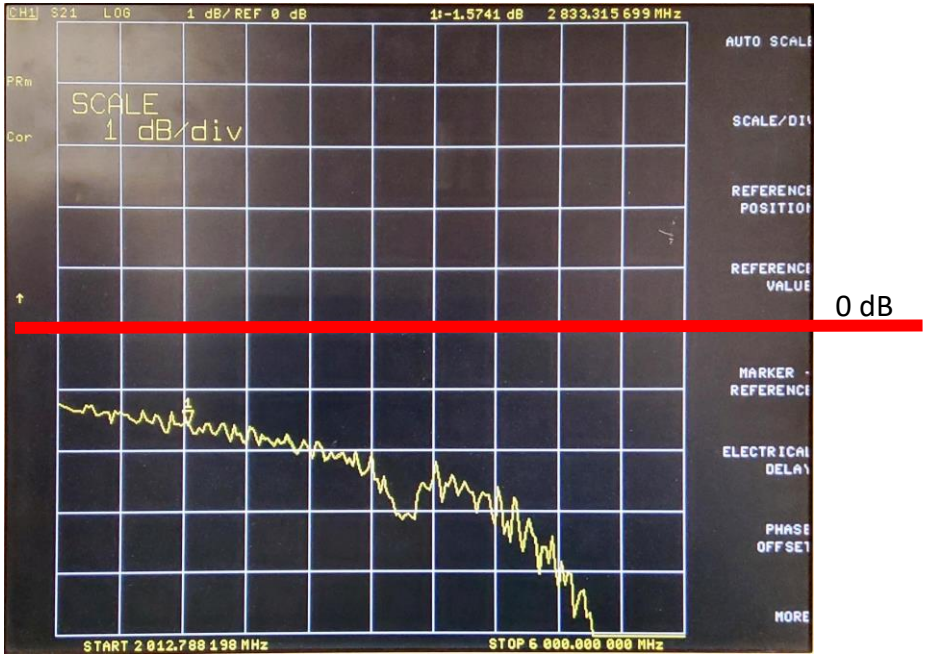
A company based in Gothenburg wants to sponsor us in conducting some RF tests and more.

Signal Loss in the cables:

By sending a 0 dB signal (=1 watt) in a cable and measuring the power of signal at the end of the cable, it will indicate how much losses the cable introduces. This test can be performed for different frequencies. Frequencies of interest are 2.4 GHz and 5 GHz because they are popular Wi-Fi frequencies standard.

Cable	@ 2.4 GHz [dB]	@ 5 GHz [dB]
In box cables	-0.5	-2.2
2 M Wi-Fi cable	-1.4	-3.2
3 M Wi-Fi cable	-2.1	-5
Base station	-0.8	-1.1

The signal loss in the cables measured at EMC Services. The "in box cables" refer to the short antenna cables that connect the router to the wall of the waterproof box inside the side-structure. The 2 and 3 M Wi-Fi cables are the cables that connect the car's antennas with the side-structures's waterproof box. The "Base station" refer to the base station's 1 M antenna cables.



The x axis shows the frequencies from 2 Ghz up 6 GHz with dev 400 MHz while the y axis shows the signal damping in the 3 m long antenna cable where each dev is equal to 1 dB.

Telemetry: Test results

- *EMC Services AB:*

VSWR / Reflection loss:

In this test, a signal of 0 dB (equivalent to 1 watt) is sent to the antenna. The Vector Network Analyzer (VNA) measures how much of the signal is reflected back from the antenna. A lower reflection loss indicates that more of the signal's energy is transmitted through the antenna.
Note: A lower reflection loss does not necessarily indicate a better-performing antenna. Even if the power is not reflected back, it could be dissipating as heat rather than radiating as an RF signal.

***Conclusion**: The reflection loss of the antennas decreases when they are placed near the side structure or the monocoque. It may be worth investigating the possibility of laminating the PCB antennas onto the monocoque or rear wing to reduce weight and improve aerodynamics. However, the radiation pattern and the EIRP should be carefully evaluated before proceeding with lamination, as we currently have no data on how the antennas will behave in these locations. The only thing we know so far is that the VSWR decreases, which is a positive sign. More investigation is needed.*

Antenna	In free space	On side structure	On monocoque
Cylindrical antennas	-6 dB	-10 dB	-12 dB
PCB antennas	-10 dB	-12 dB	-12 dB
Base station	-11 dB	-	-

The table summarizes the power reflected back from the antennas in different situations. This is based on the graphs on the next slide, which may be difficult to read due to their poor quality.

Telemetry: Test results

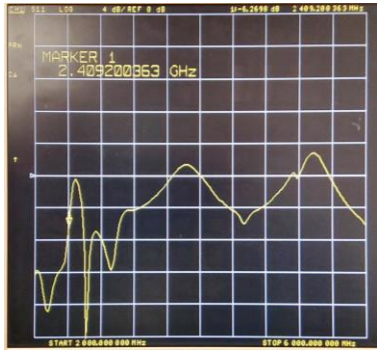
- *EMC Services AB:*

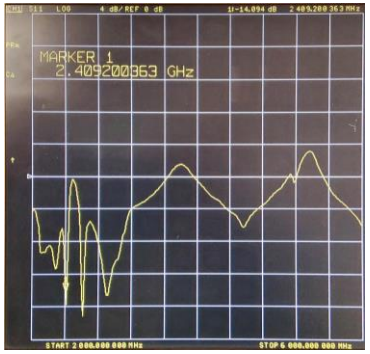
VSWR / Reflection loss:

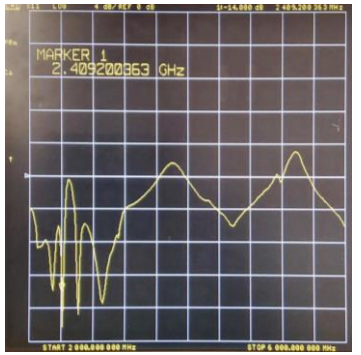
The graphs here show how much reflection was received from the transmitted 0 dB signal. The quality of the graphs is poor because they were captured using an old VNA, which made saving the graphs difficult. As a result, I had to take a picture of its screen. Note: each dev in the graphs corresponds to 10 dB.

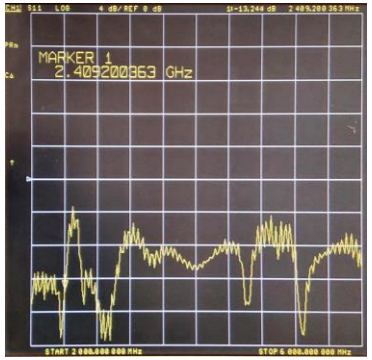
- *Cylindrical antennas*

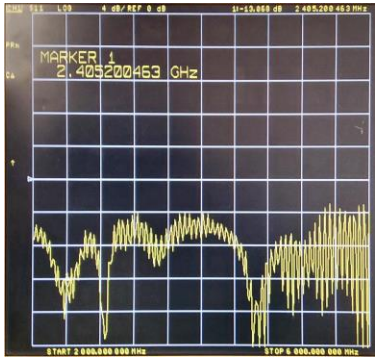
- *PCB antennas*

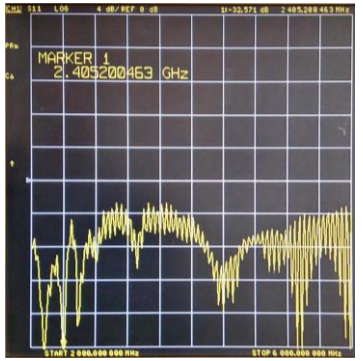












In free space

On side-structure

On monocoque

Telemetry: Test results

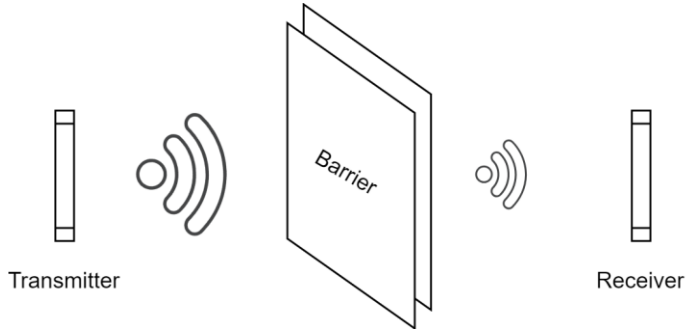
- *EMC Services AB:*

Penetration loss:

The test was performed as follows:

A transmitter antenna sends an RF signal through the air, which is received by a receiver antenna. The power of the received signal is used as a reference. Next, a barrier is placed between the transmitter and the receiver, and the power of the received signal is compared with the reference signal power. The difference between the two signals indicates how much loss the barrier introduces.

In our case, I was interested in evaluating the impact of placing the antennas inside the side structure or inside the monocoque. So, the barriers I used were: 1- Side structure test-panel. 2- Monocoque test-panel.



The results were as follows:

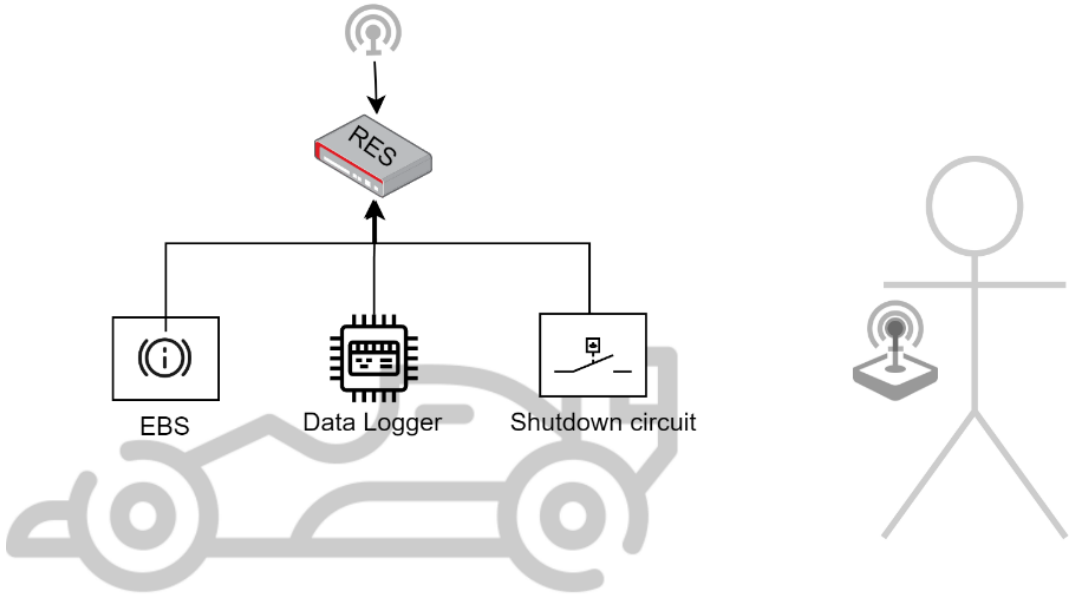
- When the antennas were placed inside the monocoque, the RF signal lost 15 dB of its original power.

- When the antennas were placed inside the side structure, the RF signal lost 10 dB of its original power.

RES: Test results

Measurements at EMC:

- *The RES vehicle module does not transmit any signal.*
- *The RES transmitter module - with the training Race E-Key- utilizes frequency 434 MHz + 0.1Mhz * selector switch.*
- *Power = 23.11 dBm.*
- *Band width: 1 MHz.*
- *Antenna cable loss =*
  - *Long cable: -4.15 dB @ 430 MHz*
  - *In box: -0.4 dB @430 MHz*



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Components I bought

- *Ethernet tester: Very useful to check manual crimped ethernet cables and even the grounding/shielding of the cables.*
- *Crystal detector: Very handy and easy to use to measure RF power,but it is not accurate.*
- *Self amalgamating tape: For waterproofing joints on cables.*
- *Coaxial connector crimper: To crimp SMA/RP-SMA connectors*
- *Telemetry Hard Case: To store and keep the base station and the RES transmitter module.*
- *Routers for the car and base station.*

Recommendation

- Upgrade the in-car router to rutx11:

To get 10 times higher transfer speed when downloading the logged data from the DVPC using the ethernet outlet under the dash.

- Case lab has a spectrum analyzer up to 3 GHz:

Use it to do measurement on the 2.4 GHz Wi-Fi, or RES RF signal. In CFS24, I bought a crystal detector which can be used to measure the amplitude of an RF signal with an oscilloscope, but it is not accurate that is why it is good to use the spectrum analyzer.

- Coverage test:

The best place I could find to perform telemetry coverage test was at Säve, 20 minutes from Chalmers. There you can find a long straight street with almost no obstacles. Here it is 57°48'09.0"N 11°53'06.6"E.

- Telemetry reliability:

The base station disconnect from the car's router specially if we turn on and off the car's router (turn on and off the whole car). A simple solution was to restart the base station.

Dual frequency antennas:

The antennas that we got from Quwireless are really good antennas and they can operate at both 2.4Ghz and 5 Ghz.

Recommendation

- Hight matters:

Mounting the cars antennas higher from ground increase the coverage drastically but disturbes the air flow in front of the rear wing and the aero subgroup will not that happy.

- Boost Wi-Fi power

Use Australia Wi-Fi sittings to boost the wifi signal comes out from teltonika routers. Note: You need to calculate the cable loss and boos the Wi-Fi to compensate for cable losses. Remember: the transmitted 2.4Ghz signal should not exceed 20 dBm in Europe.

- Waterproof testing:

The way we tested the waterproofing of our enclosures was really bad. We placed some tissues inside the enclosures and the place them under the water for a couple of seconds and then open them and check if tissues get wet or not. A better approach is to close the box and connect it to an air pump to create a positive pressure inside it and then spray soapy water on the box. If there is any leakage you will see bubbles immediately.

- CAN Selector Switch:

As described it is not needed anymore, It is up for you to keep it or remove it for an easier debugging and installation of the CAN buses.

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If you have any questions or would like to discuss further, please don't hesitate to contact me.

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aymanalzein@hotmail.com

Good Luck!